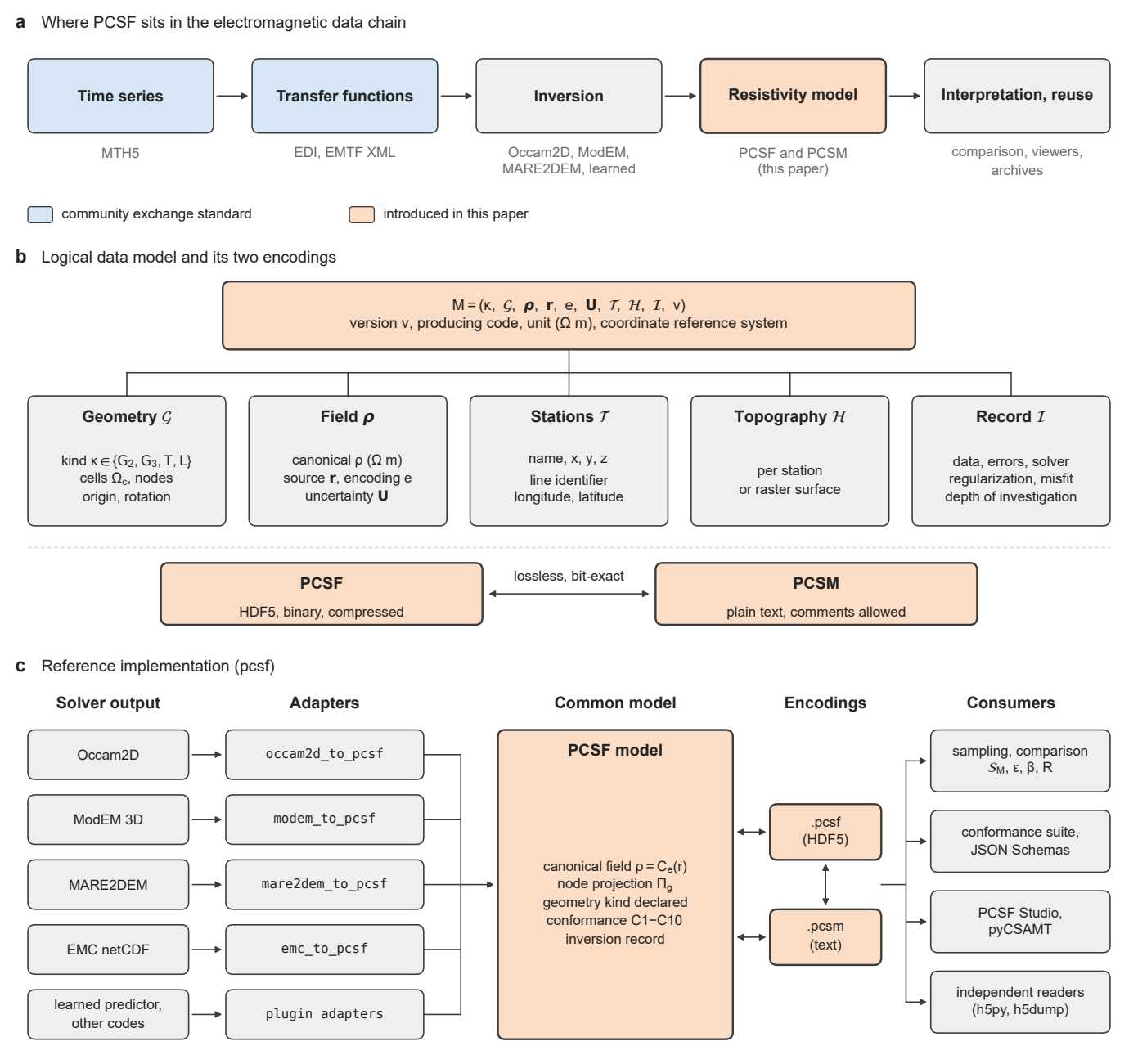

a Where PCSF sits in the electromagnetic data chain
Time series
MTH5
Transfer functions
EDI, EMTF XML
Inversion
Occam2D, ModEM, MARE2DEM, learned
Resistivity model
PCSF and PCSM (this paper)
Interpretation, reuse
comparison, viewers, archives
community exchange standard
introduced in this paper
b Logical data model and its two encodings
M = (κ, 𝒢, ρ, r, e, U, 𝒯, ℋ, ℐ, v)
version v, producing code, unit (Ω m), coordinate reference system
Geometry 𝒢
kind κ ∈ {G2, G3, T, L}
cells Ωc, nodes
origin, rotation
Field ρ
canonical ρ (Ω m)
source r, encoding e
uncertainty U
Stations 𝒯
name, x, y, z
line identifier
longitude, latitude
Topography ℋ
per station
or raster surface
Record ℐ
data, errors, solver
regularization, misfit
depth of investigation
PCSF
HDF5, binary, compressed
lossless, bit-exact
PCSM
plain text, comments allowed
c Reference implementation (pcsf)
Solver output
Adapters
Common model
Encodings
Consumers
Occam2D
ModEM 3D
MARE2DEM
EMC netCDF
learned predictor, other codes
occam2d_to_pcsf
modem_to_pcsf
mare2dem_to_pcsf
emc_to_pcsf
plugin adapters
PCSF model
canonical field ρ = Ce(r)
node projection Πg
geometry kind declared
conformance C1−C10
inversion record
.pcsf (HDF5)
.pcsm (text)
sampling, comparison SM, ε, β, R
conformance suite, JSON Schemas
PCSF Studio, pyCSAMT
independent readers (h5py, h5dump)